Simulation Assumptions Schedule

Project: Miracle Lake, Karnes County, South Texas — Phase 1 **Model:** OCTO-optimized digital-twin simulation
Timestamp: 2026-07-06T19:53:22.251Z
Dashboard: https://octo-dashboard-rikj.onrender.com/
Status: simulation-validated — live certification begins at Phase 1 energisation

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1. Site and hardware

Assumption	Value	Source
Site	Miracle Lake, Karnes County, South Texas — Phase 1	Project definition (Xplor Energy / Karnes Electric)
Containers	3 × Bitmain ANTSPACE HK3 V6 (70 miner slots each)	Site config
Miners	210 × Antminer S23 Hyd 580TH	asicminervalue.com (retrieved 2026-07-05)
Nominal hashrate	121.8 PH/s	210 × 580 TH
Nominal power draw	1,157.1 kW	Fleet nameplate
Fleet efficiency	9.5 J/TH	Miner spec sheet

2. Power and network economics

Assumption	Value	Source / logic chain
Electricity rate	**$0.045/kWh**	Karnes County TX contract — Xplor Energy / Karnes Electric
BTC price (model)	**$63,581**	Input at 2026-07-06T19:53:22.251Z
Block reward	3.125 BTC	Post-2024 halving
Blocks per day	144	Bitcoin protocol
Network difficulty	Modeled at pull	2026-07-06T19:53:22.251Z twin run

3. Scenario comparison (baseline vs OCTO-optimized)

Parameter	Industry-typical baseline	OCTO-optimized digital-twin
Control layer	Rig-level SCADA, no OCTO skill	octo-mining per-chip thermal co-control
Hashrate utilization	**87.9%**	**100%**
Utilization basis	CleanSpark Feb 2026 operating/peak 86.4%; model uses mid-upper range	Recovered hashrate from thermal throttling
Hashrate gain vs baseline	—	**+13.7%**
Efficiency gain (J/TH)	—	**−12.1%**
Daily profit	Baseline scenario in twin	**$2,387/day**
Edge vs baseline	—	**+22.6%**

Industry-typical baseline means rig-level SCADA without OCTO per-chip thermal co-control, modeled at **87.9% hashrate utilization** (within CleanSpark Feb 2026 operating/peak range); **Edge vs baseline** is the simulated daily-profit uplift of the OCTO-optimized digital-twin scenario over that unoptimized baseline.

4. Loan-sizing methodology

Step	Method
Tier daily profit	OCTO-optimized $2,387/day × (miners funded ÷ 210)
Tier monthly profit	Daily profit × (365 ÷ 12)
IO payment	Principal × 12% ÷ 12
Post-IO payment	Tier monthly profit ÷ 1.3x target DSCR
Balloon	Principal remaining after months 7–48 at target-DSCR payments
Build-out sizing	Locked CAPEX quotes (Schedule C) applied to draw amount

5. Data room

The **full OCTO-optimized digital-twin economics model** — including network difficulty sensitivity, power-cost scenarios, and container/miner deployment logic — is available in the lender data room. This schedule summarizes inputs material to debt sizing; definitive diligence should reference the complete model export.

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